

AP CALCULUS AB – UNIT 2

2.2 Tangent Lines and the Derivative



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Represent a derivative at a point and a derivative function as limits of difference quotients.
- Translate among $f'(a)$, y' , $\frac{dy}{dx}$, and related notation.
- Use a derivative value to write the equation of a tangent line.

Keys: Essential Knowledge and Vocabulary

Derivative at a point

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}, \text{ provided the limit exists.}$$

Derivative function

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Equivalent form

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

Tangent line

$$y - f(a) = f'(a)(x - a).$$

Explanation, Strategy, and Common Errors

- The base point must remain fixed in a derivative-at-a-point limit. In the h -form, the nearby input is $a + h$; in the $x \rightarrow a$ form, the denominator is $x - a$.
- Notation communicates both the function and the variable of differentiation. For example, $\frac{dP}{dt}$ is the rate of change of P with respect to t .
- A derivative value supplies only the tangent slope. To write the tangent line, also use the point $(a, f(a))$.

Solved Example: Definition at a point

Use the limit definition to find $f'(3)$ for $f(x) = x^2 - 4x$.

Answer and Reasoning

$$\frac{(3+h)^2 - 4(3+h) - (-3)}{h} = \frac{2h+h^2}{h} = 2 + h, \text{ so } f'(3) = 2.$$

Solved Example: Recognizing a derivative

Interpret $\lim_{x \rightarrow 2} \frac{\ln x - \ln 2}{x - 2}$.

Answer and Reasoning

It is the derivative of $f(x) = \ln x$ at $x = 2$, so its value is $f'(2) = \frac{1}{2}$.

Solved Example: Tangent equation

Given $g(4) = -1$ and $g'(4) = 3.5$, write the tangent line.

Answer and Reasoning

$$y + 1 = 3.5(x - 4).$$

AP Exam Language

For every derivative in context, include the sign, units, input value, and a complete verbal interpretation. For every existence claim, state the relevant hypothesis or one-sided comparison.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. Write $f'(5)$ using the h -form of the derivative definition.

2. Write $g'(a)$ using a variable x approaching a .

3. If $y = x^3$, state three valid notations for its derivative.

4. Given $f(2) = 7$ and $f'(2) = -4$, write the tangent line at $x = 2$.

5. Identify the function and base point in $\lim_{h \rightarrow 0} \frac{\sqrt{9+h}-3}{h}$.

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

6. Use the definition to find $f'(1)$ for $f(x) = x^2 + 3x$.

7. Use the definition to find $g'(4)$ for $g(x) = \sqrt{x}$.

8. Use the definition to derive the derivative function of $f(x) = 5x - 7$.

9. Given $p(1) = 6$ and tangent line $y = 2x + 4$, find $p'(1)$.

10. Express $\left. \frac{dV}{dr} \right|_{r=3}$ as a limit using h .

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

11. For $f(x) = \frac{1}{x}$, use the definition to derive $f'(x)$ for $x \neq 0$.

12. A limit is $\lim_{h \rightarrow 0} \frac{(2+h)^5 - 32}{h}$. Identify it and evaluate.

13. Given $f(a) = b$ and $f'(a) = m$, show that $\lim_{x \rightarrow a} \frac{f(x) - b - m(x-a)}{x-a} = 0$.

14. The tangent line to f at $x = 3$ passes through $(3, 5)$ and $(7, -1)$. Find $f(3)$ and $f'(3)$.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

15. Use the definition to find the derivative function of $f(x) = x^2 - 2x + 4$.

16. Evaluate $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4}$ by interpreting it as a derivative.

17. Suppose $f(2) = 3$ and $\lim_{x \rightarrow 2} \frac{5f(x)-15}{x-2} = 20$. Find $f'(2)$.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

18. If $f'(a)$ exists, prove $\lim_{h \rightarrow 0} \frac{f(a+2h)-f(a)}{h} = 2f'(a)$.

19. For $f(x) = x|x|$, use the definition to determine $f'(0)$ even though the formula changes at 0.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. Which limit equals $f'(4)$?

(A) $\lim_{h \rightarrow 0} \frac{f(4+h)-f(h)}{4}$

(B) $\lim_{h \rightarrow 0} \frac{f(4+h)-f(4)}{h}$

(C) $\lim_{h \rightarrow 4} \frac{f(h)-f(4)}{h}$

(D) $\lim_{x \rightarrow 0} \frac{f(x)-f(4)}{x-4}$

2. $\lim_{x \rightarrow 3} \frac{x^4-81}{x-3}$ represents

(A) the value of x^4 at 3

(B) the derivative of x^4 at 3

(C) the average rate on $[0, 3]$

(D) the derivative of x^3 at 4

3. If $f(1) = 2$ and $f'(1) = -3$, the tangent line is

(A) $y - 1 = -3(x - 2)$

(B) $y - 2 = -3(x - 1)$

(C) $y + 2 = 3(x + 1)$

(D) $y - 2 = x - 3$

4. Which notation explicitly identifies the independent variable?

(A) f'

(B) y'

(C) $\frac{dy}{dx}$

(D) D

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. For $f(x) = e^x$, a calculator approximation of $\frac{f(1.001)-f(1)}{0.001}$ is closest to

(A) 1

(B) 2.718

(C) 7.389

(D) 0

6. A tangent-line calculator reports $y = 4.237x - 8.912$ at $x = 2.4$. Which value estimates $f'(2.4)$?

(A) -8.912

(B) 2.4

(C) 4.237

(D) 1.257

AP-Style Free-Response Practice

Calculator Permitted

Complete the following independent calculator-permitted problems. Show the mathematical setup and justify conclusions when requested.

(1a) Problem 1: Derivative Definition. Let $f(x) = \frac{2}{x+1}$.

Write $f'(a)$ as a limit in h .

[2 points]

(1b) Algebraically simplify the difference quotient.

[3 points]

(1c) Determine $f'(a)$ and write the tangent line at $a = 1$.

[3 points]

No Calculator

Complete the following independent non-calculator problems. Show the mathematical setup and justify conclusions when requested.

(1a) Problem 1: Derivative Definition. A differentiable function q satisfies $q(3) = 8$. A calculator gives $q(3.001) = 8.004997$ and $q(2.999) = 7.995003$.

Estimate $q'(3)$ with a symmetric difference.

[2 points]

(1b) Write an equation of the estimated tangent line at $x = 3$.

[2 points]

(1c) Use the line to estimate $q(3.04)$.

[2 points]

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